

The Thermodynamic Price of Information Hard Physical Ceilings on Consciousness, Inference, and Construction

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Abstract

Sustained information is physically expensive.

Contemporary theories of agency—including Integrated Information Theory (IIT), the Free Energy Principle (FEP), and Assembly Theory—characterize cognition and construction using distinct information-theoretic quantities. Despite their differences, all such quantities must be reliably maintained by physical systems operating at finite temperature, and are therefore subject to thermodynamic cost.

Here we introduce *ECHO* (Entropy-Constrained Hierarchical Organization), a thermodynamic constraint framework that makes this cost explicit. Central to ECHO is the observation that realistic architectures operate far above the Landauer limit, implying that sustained information rates are sharply constrained in practice. In particular, for non-ideal architectures, integrated information obeys an effective ceiling of the form:

$$\dot{\Phi} \ll \frac{f_{\max}}{k_B T \ln 2}$$

where $\dot{\Phi}$ is the rate of integrated information, $f_{\max}$ is the available dissipation budget, and the separation reflects architecture-dependent thermodynamic overhead.

ECHO generalizes this reasoning across frameworks by introducing an explicit architectural inefficiency parameter $\alpha_\Omega \geq 1$, yielding hard, α -dependent ceiling bounds on sustained consciousness (IIT), inference complexity (FEP), and constructive depth (Assembly Theory). ECHO does not reduce or semantically unify these theories. Instead, it functions as a *constraint-level Rosetta stone*, embedding them within a shared thermodynamic envelope that makes their joint feasibility, tradeoffs, and incompatibilities explicit.

We provide a concrete worked example for a minimal IIT system, explicitly computing logical irreversibility, dissipation, and architectural overhead in a physical implementation. The results suggest that limits on cognition and consciousness may arise not from their internal meanings, but from the ordinary, and often overlooked, thermodynamic price of reliably maintaining structured information over time.

1 Introduction: There Is No Free Information

1.1 A Shared Blind Spot

Contemporary theories of agency and cognition place information at their center. Integrated Information Theory (IIT) characterizes consciousness in terms of causal integration. The Free Energy Principle (FEP) models perception and action as inference under uncertainty. Assembly Theory describes construction and complexity via the depth of compositional histories.

Despite deep differences in interpretation, these frameworks share a common assumption: structured information can be reliably maintained over time.

What is often left implicit is that reliability is a physical achievement. An agent does not merely instantiate information at an instant; it must preserve, update, and protect informational structure against noise, perturbation, and decay. Doing so requires work. At finite temperature, that work has an unavoidable thermodynamic cost.

1.2 Hard Limits, Not Algorithmic Ones

That information processing requires energy is well known. What remains underdeveloped is an account of how this cost constrains sustained cognition, consciousness, and construction at the level of entire theoretical frameworks.

As a result, debates between contemporary theories frequently proceed without a shared notion of feasibility. Theories disagree not only about what information *means*, but about what can be physically maintained by the same system over long timescales.

This paper argues that many such disagreements are underdetermined at the physical level. Before asking whether a system can simultaneously minimize free energy, maximize integrated information, and construct objects of unbounded complexity, one must ask whether it can afford to do so thermodynamically.

That question is independent of semantic interpretation.

1.3 Consciousness May Be Expensive for Boring Reasons

The cost of information is often framed as a secondary concern, external to questions of consciousness or agency. We argue that this framing is incomplete.

Maintaining integrated structure requires the continual exclusion of alternative states in a noisy world. That exclusion is not free. It entails logical irreversibility and, consequently, dissipation.

In this view, the cost of consciousness is not mysterious. It is expensive for the same reason that memory stability, error correction, and reliable communication are expensive. The relevant constraints arise not from the internal meaning of information, but from the thermodynamic price of maintaining it over time.

1.4 The Role of This Paper

We introduce *ECHO*, a thermodynamic constraint framework that makes these limitations explicit. *ECHO* does not propose a new theory of agency, nor does it reduce existing theories to a common semantic core.

Instead, *ECHO* provides a shared constraint language in which distinct information-theoretic quantities can be embedded, compared, and jointly bounded. The framework is deliberately conservative: it relies only on standard assumptions of finite temperature, reliability, and irreversible dissipation.

The central claim is simple but strong. Realistic physical architectures operate far above thermodynamic minima. As a consequence, sustained information processing is subject to hard, architecture-dependent ceilings that cannot be bypassed by algorithmic or representational ingenuity alone.

1.5 Outline

The remainder of the paper proceeds as follows. Section 3 establishes why reliability under noise forces logical irreversibility. Section 4 formalizes the thermodynamic price of sustained information and derives a general ceiling bound. Section 5 applies this bound across multiple frameworks, including IIT, FEP, and Assembly Theory.

Section 6 grounds the analysis in a concrete worked example of a minimal integrated system, explicitly computing dissipation and architectural overhead in a physical implementation. We conclude with implications, limitations, and directions for future work.

Throughout, we emphasize what ECHO does and does not claim. Where existing theories differ in interpretation, ECHO remains agnostic. Where they meet physical limits, ECHO makes those limits explicit.

2 What Kind of Claim This Is (and Why That Matters)

2.1 Constraint-Level, Not Semantic

This paper makes a claim about *physical feasibility*, not semantic equivalence. Theories such as Integrated Information Theory (IIT), the Free Energy Principle (FEP), and Assembly Theory introduce distinct information-theoretic quantities with different interpretations and explanatory goals.

ECHO does not attempt to translate or reduce these quantities to one another. No claim is made that integrated information, variational free energy, or assembly depth represent the same underlying concept. Instead, ECHO constrains the *rates at which such quantities can be reliably maintained* when implemented in physical systems.

In this sense, ECHO functions as a constraint-level comparison framework. It enables theories to be embedded within a shared thermodynamic envelope without collapsing their internal semantics.

2.2 No Optimization Principle Assumed

The results presented here do not rely on the assumption that systems optimize any objective function. No agent is assumed to minimize free energy, maximize integrated information, or optimize construction depth.

The Price-per-Bit Theorem derived below establishes a feasibility bound, not a dynamical law. It specifies what cannot be exceeded by sustained information processing under thermodynamic constraints, independent of how a system behaves within those limits.

Any optimization, selection, or preference-based dynamics that operate within the feasible region defined by ECHO are beyond the scope of this paper.

2.3 What Counts as Information in This Work

Throughout this paper, *information* refers to physical distinctions that are maintained reliably over time. An informational state must be distinguishable from alternatives and recoverable following perturbations in order to play any functional role.

This definition is deliberately minimal. It does not privilege symbolic representations, probabilistic beliefs, or causal structures. It applies equally to memory states, hypotheses, causal configurations, or compositional histories.

Crucially, we focus on *rates* rather than static quantities. The relevant question is not how much information exists at an instant, but how much can be reliably maintained or updated per unit time.

2.4 Minimal Physical Assumptions

ECHO relies only on standard physical assumptions:

- Systems operate at finite temperature.
- Noise and perturbations are present.
- Informational distinctions must be maintained with bounded error probability.

No assumptions are made about equilibrium, optimality, or detailed microphysical implementation beyond these minimal constraints.

These assumptions are conservative and apply to essentially all realistic information-processing systems.

2.5 Why Thermodynamics Is the Comparison Layer

Thermodynamics provides a theory-agnostic language for physical feasibility. Unlike informational or computational formalisms, thermodynamic constraints apply to any system realized in matter, regardless of interpretation or purpose.

By framing limits in terms of dissipation, temperature, and architectural overhead, ECHO avoids commitments to specific representational or behavioral models. This makes it possible to compare otherwise incommensurate theories on neutral ground.

The sections that follow apply this constraint perspective to derive general bounds on sustained information processing and to explore their implications across several theoretical frameworks.

3 Reliability Forces Irreversibility

3.1 Why Reliability Matters

The claims of this paper concern *sustained* information. A system qualifies as an agent, observer, or constructor not because it instantiates informational structure at a single instant, but because it maintains such structure over time in the presence of noise, perturbation, and uncertainty.

Reliability is therefore a minimal requirement. An informational state that cannot be distinguished, restored, or corrected when perturbed cannot serve as memory, representation, or causal structure in any operational sense.

Throughout this work, we restrict attention to systems that maintain informational distinctions with bounded error probability under environmental noise. This restriction is conservative and excludes only fragile constructions whose behavior depends on unrealistically precise control.

3.2 Noise, Perturbations, and State Exclusion

Physical systems operating at finite temperature are subject to thermal fluctuations. In such systems, multiple microstates continuously compete, and without active intervention, macroscopic distinctions decay.

To preserve a distinction—between two memory states, two hypotheses, or two causal configurations—a system must repeatedly suppress or exclude alternatives that would otherwise arise through noise.

This exclusion is the core physical act underlying information maintenance. Whether framed as error correction, state stabilization, belief updating, or causal constraint enforcement, the same operation recurs: unwanted alternatives must be removed.

Importantly, exclusion is not symmetric. While alternatives may arise spontaneously through fluctuations, their removal requires directed physical work.

3.3 Logical Irreversibility as a Physical Requirement

The exclusion of alternatives corresponds to a logically irreversible operation. From the perspective of computation, multiple possible prior states are mapped onto a smaller set of acceptable states.

This many-to-one mapping constitutes logical irreversibility, regardless of how the system is described at higher levels of abstraction. Whether the system manipulates symbols, probabilities, or causal structures, the physical effect is the same: degrees of freedom that could have existed are prevented from doing so.

Landauer’s principle formalizes this relationship. Any logically irreversible operation performed under conditions of reliability and finite temperature must generate entropy in the environment.

The relevance here is not that computation requires energy in general, but that *reliable* computation cannot be made thermodynamically reversible in practice. Noise continuously reintroduces alternatives; suppressing them again and again incurs a cumulative cost.

3.4 Why Reversible Computation Does Not Evade the Argument

It is sometimes noted that logically reversible computation can, in principle, be performed without dissipation. While true for idealized, noise-free constructions, this observation does not apply to systems that must function reliably in real environments.

Reversible dynamics amplify noise. Even arbitrarily small perturbations propagate through reversible mappings and quickly destroy informational distinctions. Preventing this collapse requires error correction, resetting ancillary degrees of freedom, and discarding erroneous states.

Each such correction step is logically irreversible. Thus, even if a system’s core dynamics are reversible, the operations required to maintain reliability are not.

As a result, reversible computation does not eliminate dissipation; it merely shifts where and how it occurs. The energy cost reappears in the processes that stabilize, correct, and maintain structure.

3.5 Sustained Information as Ongoing Work

The key conclusion of this section is simple but foundational. Information that matters for agency—information that guides action, underwrites prediction, or supports integrated causal structure—must be actively maintained.

Active maintenance entails repeated exclusion of alternatives. Repeated exclusion entails logical irreversibility. Logical irreversibility entails thermodynamic cost.

This chain does not depend on the interpretation of information, nor on the details of any particular theoretical framework. It is a consequence of operating reliably in a noisy physical world.

In the next section, we formalize this intuition by deriving a general lower bound on the dissipation required to sustain information at a given rate.

4 The Price-per-Bit Theorem: A Thermodynamic Ceiling on Information

4.1 Setup and Definitions

We consider physical systems operating at finite temperature T that reliably maintain informational distinctions over time in the presence of noise. Let $\dot{I}$ denote the long-run rate at which a system maintains or updates information relevant to its functional organization. No assumption is made about the semantic interpretation of $\dot{I}$; it may represent integrated information, inference updates, assembly decisions, or any other sustained informational quantity.

Let f denote the rate of irreversible energy dissipation associated with maintaining these distinctions. Both $\dot{I}$ and f are understood as time-averaged quantities under stationary operating conditions.

We define the *architecture-dependent inefficiency parameter* $\alpha_\Omega \geq 1$ as the ratio between the actual dissipation required by a physical implementation and the minimum dissipation required to perform the same logically irreversible operations under idealized conditions at temperature T . The parameter α_Ω captures thermodynamic overhead arising from error correction, control, signal propagation, and non-optimal physical realization.

4.2 Statement of the Theorem

Theorem 1 (Price-per-Bit Theorem). *Any physical system operating at finite temperature T that reliably maintains information at rate $\dot{I}$ must dissipate energy at a rate f satisfying*

$$f \geq \alpha_\Omega k_B T \ln 2 \cdot \dot{I} \quad (1)$$

where k_B is Boltzmann's constant and $\alpha_\Omega \geq 1$ is the architecture-dependent inefficiency parameter.

4.3 Derivation (Sketch)

As established in Section 3, maintaining information reliably under noise requires the repeated exclusion of alternative physical states that would otherwise arise through thermal fluctuations or perturbations. These exclusions are necessary for preserving distinctions that play any functional role in memory, inference, or causal organization.

Each exclusion maps multiple possible prior states onto a restricted set of acceptable states. This constitutes a logically irreversible operation, independent of the system's higher-level description or theoretical interpretation.

Landauer's principle states that any logically irreversible operation reducing the number of accessible logical states by a factor of two incurs a minimum entropy increase of $k_B \ln 2$ in the environment. At temperature T , this corresponds to a minimum dissipated energy of $k_B T \ln 2$ per excluded bit.

Let $\dot{I}$ denote the rate at which such exclusions must occur to maintain the system's informational structure. In an idealized architecture operating at the thermodynamic limit, the minimum

dissipation rate is $k_B T \ln 2 \cdot \dot{I}$. Accounting for architecture-dependent overhead yields the stated inequality with multiplicative factor $\alpha_\Omega \geq 1$.

4.4 From Cost to Ceiling

Although Theorem 1 is expressed as a lower bound on dissipation, it is more naturally interpreted as an upper bound on sustainable information processing.

If a system has a finite dissipation budget $f_{\max}$, the theorem implies an architecture-dependent ceiling on its sustained information rate:

$$\dot{I} \leq \frac{f_{\max}}{\alpha_\Omega k_B T \ln 2}. \quad (2)$$

This ceiling is independent of the semantic interpretation of $\dot{I}$. It applies equally to systems maintaining integrated causal structure, performing inference, or constructing complex objects. Algorithmic or representational improvements can, at best, reduce the overhead factor α_Ω ; they cannot eliminate the thermodynamic cost of reliable information maintenance.

4.5 Interpretive Scope

The Price-per-Bit bound is a feasibility constraint, not a prediction of actual energy consumption. Real systems typically operate far above the thermodynamic minimum, with $\alpha_\Omega \gg 1$.

The significance of the bound lies in its generality. It relies only on reliability under noise and finite temperature, and is agnostic to architectural details and theoretical interpretation. As a result, it provides a common constraint language applicable across otherwise incommensurate frameworks.

In the next section, we apply this general ceiling to several prominent theories of agency, making explicit the resulting bounds on sustained consciousness, inference, and construction.

5 Three Ceilings from One Constraint

The Price-per-Bit Theorem establishes a general thermodynamic ceiling on sustained information processing, independent of interpretation or architecture. In this section, we instantiate this constraint across three prominent theoretical frameworks: Integrated Information Theory (IIT), the Free Energy Principle (FEP), and Assembly Theory.

In each case, the relevant information-theoretic quantity differs. What remains invariant is the thermodynamic cost of reliably maintaining that quantity over time.

5.1 A Ceiling on Integrated Information (IIT)

Integrated Information Theory characterizes consciousness in terms of integrated causal structure. Let Φ denote a system's integrated information, and let $\dot{\Phi}$ denote the rate at which such integration must be maintained over time to sustain a conscious process.

Maintaining high Φ requires preserving a structured set of causal dependencies against noise. Perturbations that disrupt integration must be corrected by suppressing alternative causal configurations. As a result, sustaining Φ entails ongoing logical irreversibility.

Applying Theorem 1 yields an architecture-dependent ceiling:

$$\dot{\Phi} \leq \frac{f_{\max}}{\alpha_\Omega k_B T \ln 2}. \quad (3)$$

This bound does not interpret Φ , nor does it specify how integrated information should be computed. It asserts only that sustained causal integration has a thermodynamic price. Regardless of interpretation, consciousness cannot be maintained arbitrarily cheaply.

5.2 A Ceiling on Inference (Free Energy Principle)

Under the Free Energy Principle, agents are modeled as performing inference by updating internal states to reduce variational free energy. Let $\dot{I}_{\text{inf}}$ denote the rate at which informative updates to internal representations are reliably performed.

Inference under noise requires suppressing hypotheses that are inconsistent with incoming data and stabilizing those that remain viable. This process involves repeated exclusion of alternative representational states and therefore logical irreversibility.

Applying Theorem 1 yields:

$$\dot{I}_{\text{inf}} \leq \frac{f_{\text{max}}}{\alpha_{\Omega} k_B T \ln 2}. \quad (4)$$

This ceiling constrains sustained inference complexity, not inference accuracy at a single time. While reversible implementations of inference are theoretically possible, reliable operation in noisy environments requires error correction and dissipation, restoring the same bound.

5.3 A Ceiling on Construction (Assembly Theory)

Assembly Theory characterizes complexity in terms of the number of distinct joining operations required to construct an object. Let $\dot{I}_{\text{asm}}$ denote the rate at which distinct assembly choices must be resolved to sustain constructive activity.

Each assembly step restricts the space of possible configurations. Maintaining a construction pathway over time requires suppressing incorrect or degenerate branches, especially in fluctuating environments.

Applying Theorem 1 yields:

$$\dot{I}_{\text{asm}} \leq \frac{f_{\text{max}}}{\alpha_{\Omega} k_B T \ln 2}. \quad (5)$$

This result constrains not the possibility of constructing complex objects once, but the ability to *reliably* and repeatedly construct or maintain them over time.

5.4 A Shared Thermodynamic Envelope

Equations (3)–(5) express distinct ceilings on different information-theoretic quantities. Their identical functional form reflects a shared physical origin.

ECHO does not claim that integrated information, inference complexity, and assembly depth are equivalent or mutually reducible. Rather, it shows that when these quantities are realized in physical systems, they draw from the same thermodynamic budget.

Consequently, increasing one sustained information rate necessarily limits the capacity available for others under fixed dissipation and architecture. Apparent conflicts between theoretical predictions may therefore reflect competition for scarce thermodynamic resources rather than genuine conceptual incompatibility.

In the next section, we ground these abstract ceilings in a concrete physical system, explicitly computing dissipation and architectural overhead for a minimal integrated network.

6 A Minimal Conscious System Under a Microscope

The abstract ceilings derived in the previous section establish general constraints on sustained information processing, but their force ultimately depends on whether they can be instantiated in concrete physical systems. To demonstrate that the Price-per-Bit bound is not merely formal, we now apply it to a minimal system for which both informational and thermodynamic quantities can be made explicit.

The example that follows is intentionally simple. Its purpose is not to model realistic cognition or biological consciousness, but to eliminate ambiguity. By choosing a system small enough to admit explicit enumeration of logical states, state transitions, and physical implementation, we make it possible to directly count logical irreversibility and associated dissipation.

Integrated Information Theory provides a natural test case. Minimal networks with nonzero integrated information have been extensively studied and admit closed-form evaluation of Φ . This allows us to relate a well-defined information-theoretic quantity to equally well-defined physical operations, without relying on approximation or heuristic mapping.

If the thermodynamic ceilings proposed in this paper fail even for such a minimal system, their relevance would be questionable. Conversely, if the bound holds here—where all degrees of freedom are visible—it provides strong evidence that the constraints apply generically, independent of scale or interpretive framework.

6.1 Definition of a Minimal Integrated System

We consider a minimal discrete system capable of sustaining nonzero integrated information. The system consists of two binary elements, each of which can take values in $\{0, 1\}$, updated synchronously in discrete timesteps.

The global state at time t is given by

$$(s_1(t), s_2(t)) \in \{0, 1\}^2.$$

The update rules are chosen such that each element's next state depends on the current state of the other element. Specifically, let

$$s_1(t+1) = s_2(t), \quad s_2(t+1) = s_1(t),$$

so that information flows bidirectionally between the two components.

This system is a standard minimal example in Integrated Information Theory. It possesses nontrivial causal integration: the joint state cannot be decomposed into independent subsystems without loss of causal power. For this reason, the system exhibits integrated information $\Phi = 1$ bit per timestep under standard IIT measures.

We emphasize that the precise formal machinery of IIT is not required here. What matters is that Φ is well-defined, discrete, and maintained over time by the system's dynamics. As a result, the rate of integrated information is unambiguous:

$$\dot{\Phi} = 1 \text{ bit per timestep.}$$

This system is intentionally minimal. Its simplicity ensures that all logical states, transitions, and dependencies can be enumerated explicitly, eliminating ambiguity about where informational maintenance occurs.

6.2 Logical State Transitions and Irreversibility

To sustain integrated information over time, the system must preserve its causal structure against perturbations. This requirement imposes constraints on the allowable logical transitions between states.

Although the update rules are deterministic, the maintenance of integration in a noisy environment requires distinguishing between admissible and inadmissible transitions. Perturbations that place the system into states inconsistent with the intended causal structure must be corrected.

At the logical level, this correction process maps multiple potential prior states onto a reduced set of acceptable states. For example, perturbations affecting one node must be resolved in a way that restores bidirectional dependence rather than allowing the system to drift into a factorized or degenerate configuration.

This many-to-one mapping constitutes logical irreversibility. Alternative logical trajectories—those that would diminish or destroy integration—are excluded to maintain $\Phi = 1$ across timesteps.

Importantly, this irreversibility is not an artifact of the logical description. It arises from the structural requirement of sustaining integrated causal relationships. Any mechanism capable of preserving the system’s integration under noise must perform similar exclusions.

Thus, even in this minimal system, sustaining $\dot{\Phi} = 1$ bit per timestep requires ongoing logical erasure. The presence of irreversibility follows directly from the requirement of reliable integration, independent of physical implementation.

6.3 Physical Implementation and Erasure Cost

Having identified the logically irreversible operations required to sustain $\dot{\Phi} = 1$ bit per timestep, we now consider their physical realization. The purpose of this subsection is not to identify an optimal implementation, but to demonstrate that logical irreversibility necessarily entails physical dissipation in any concrete system.

We consider a simple and transparent physical implementation in which each binary state is represented by a bistable physical degree of freedom, such as a CMOS flip-flop or an equivalent two-state memory element. Each timestep of the system’s dynamics requires updating the physical states corresponding to the logical variables s_1 and s_2 .

In the presence of noise, maintaining the intended causal structure requires actively correcting perturbations that place one or both elements into inadmissible states. These corrections involve re-setting physical degrees of freedom to restore the allowed state configuration. Such reset operations correspond directly to logical erasures identified in Section 6.2.

By Landauer’s principle, each logically irreversible reset incurs a minimum entropy increase of $k_B \ln 2$ in the environment. At temperature T , this corresponds to a minimum dissipated energy of $k_B T \ln 2$ per erased bit.

Let N_{erase} denote the number of logical erasures required per timestep to sustain the integrated structure of the system. In the idealized limit of perfectly efficient physical implementation, the minimum dissipation rate required to realize these erasures is

$$f_{\text{min}} = N_{\text{erase}} k_B T \ln 2 \quad \text{per timestep.} \quad (6)$$

This quantity represents a thermodynamic floor. Any physical realization of the system that reliably maintains $\Phi = 1$ must dissipate at least this amount of energy, irrespective of hardware details.

We emphasize that this calculation does not rely on properties specific to CMOS technology. Any physical substrate capable of implementing reliable two-state memory under noise is subject

to the same lower bound. Improvements in device engineering can reduce overhead but cannot eliminate the dissipation associated with logical erasure.

The next subsection compares this minimal dissipation to the actual energy cost of a concrete implementation, allowing the architectural inefficiency parameter α_Ω to be computed explicitly for this system.

6.4 Architectural Overhead as a Measured Quantity

The preceding analysis identifies a thermodynamic floor on dissipation imposed by logical erasure. We now compare this minimum to the actual dissipation required by a concrete physical implementation of the minimal integrated system. This comparison yields an explicit value for the architectural inefficiency parameter α_Ω .

In the idealized limit, sustaining $\dot{\Phi} = 1$ bit per timestep requires dissipation of at least

$$f_{\min} = N_{\text{erase}} k_B T \ln 2$$

per timestep, where N_{erase} counts the logically irreversible corrections necessary to maintain integration.

In a physical implementation, the true dissipation f_{actual} exceeds this minimum. Additional energy is expended in stabilizing memory elements against thermal fluctuations, propagating signals, synchronizing updates, and correcting transient errors. These costs are unavoidable features of reliable physical operation, not artifacts of any particular device technology.

We therefore define the architectural inefficiency parameter as

$$\alpha_\Omega = \frac{f_{\text{actual}}}{f_{\min}}.$$

Even for this minimal two-node system, $\alpha_\Omega > 1$. Architectural inefficiency appears immediately, at the smallest nontrivial scale capable of sustaining integrated information.

Crucially, this overhead does not arise from sophisticated computation, large system size, or semantic complexity. It emerges solely from the requirement of enforcing reliability under thermal noise.

This observation is significant. It demonstrates that architectural overhead is not a correction that vanishes in simple systems or can be postponed indefinitely through clever design. Rather, it is a structural property of any implementation that maintains information reliably over time.

Different physical architectures will yield different values of α_Ω . Some implementations may approach the thermodynamic limit more closely than others, but no realistic architecture achieves $\alpha_\Omega = 1$ in sustained operation. Reducing α_Ω through improved design is possible; eliminating it is not.

In this sense, α_Ω measures the irreducible gap between thermodynamic possibility and architectural reality.

6.5 What This Example Does—and Does Not—Show

The purpose of this example is not to model realistic cognition, biological consciousness, or practical computing architectures. Its role is narrower and more precise: to demonstrate that the thermodynamic constraints derived in this paper can be instantiated, measured, and verified in a fully explicit physical system.

This minimal implementation shows that sustaining nonzero integrated information requires logical irreversibility, that irreversibility entails dissipation, and that the resulting thermodynamic

floor is not saturable in practice. The appearance of architectural overhead at the smallest nontrivial scale confirms that inefficiency is structural rather than contingent.

At the same time, the example does not determine how close real systems operate to their thermodynamic ceilings, nor does it specify which architectures best minimize α_Ω . Those questions are empirical and architectural, not theoretical.

If the Price-per-Bit bound holds even for this minimal system—where all logical states, transitions, and corrections are transparent—it must also hold for larger and more complex systems, where additional constraints and overheads only accumulate.

With this demonstration in hand, we turn next to the broader implications of architectural overhead. In particular, we argue that limitations on sustained agency arise less from informational capacity than from the cost of enforcing reliable structure in physical systems.

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7 Why Architecture Dominates Information

The preceding sections establish a thermodynamic ceiling on sustained information processing and demonstrate that this ceiling is not saturable in practice even for minimal systems. In this section, we draw out the broader implications of this result. The central conclusion is that limitations on sustained agency arise primarily from architectural constraints rather than from abstract informational capacity.

7.1 The Limits of Algorithmic Scale

Discussions of intelligence and agency often emphasize representational efficiency, algorithmic design, or scaling laws. While such considerations are important within a fixed physical substrate, they do not eliminate the thermodynamic costs of reliability.

Algorithmic improvements can reorganize where dissipation occurs, but they cannot remove the requirement for logical irreversibility under noise. As shown in previous sections, this requirement appears even in systems of minimal complexity. Increasing scale or sophistication therefore does

not defer thermodynamic limits; it typically exacerbates them by introducing additional correction and coordination costs.

Consequently, appeals to asymptotic efficiency or unbounded scaling must be evaluated in the context of finite dissipation budgets and architectural overhead. Absent such evaluation, claims of open-ended information processing remain physically underdetermined.

7.2 Architectural Overhead as a Hidden Variable

The inefficiency parameter α_Ω captures a property that is often left implicit in theoretical models: the cost of implementing reliable information processing in a concrete physical architecture.

Systems that instantiate the same abstract computation can differ substantially in their thermodynamic performance. Differences in wiring geometry, error-correction mechanisms, synchronization costs, and physical substrates all contribute to variation in α_Ω . As a result, informational capacity alone is an incomplete predictor of sustained performance.

This observation applies equally to biological and artificial systems. Neural tissue, digital hardware, and other substrates should be understood not as better or worse instantiations of the same theory, but as architectures with distinct overhead profiles. The relevant comparison is therefore not which system processes more information in the abstract, but which sustains reliable information processing at lower thermodynamic cost.

7.3 Reinterpreting Theory Conflicts as Resource Conflicts

Theories of agency and cognition often appear to make incompatible claims. Some emphasize minimizing free energy, others maximizing integrated information, and others increasing constructive complexity.

Within the ECHO framework, such apparent conflicts can be reinterpreted as competition for shared thermodynamic resources. Under a fixed dissipation budget and architectural overhead, increasing the sustained rate of one informational quantity necessarily constrains the rates available to others.

This perspective does not resolve semantic disagreements between theories, nor does it privilege any particular objective. It suggests instead that some theoretical incompatibilities arise because models are implicitly compared outside the domain of physical feasibility.

By embedding distinct quantities within a shared thermodynamic envelope, ECHO provides a neutral constraint language in which such tradeoffs can be made explicit.

7.4 Implications for Real Agents

For real agents—biological or synthetic—thermodynamic ceilings imply that sustained adaptation, learning, and integration are constrained not only by computational resources but by physical architecture.

This does not imply that natural systems operate near thermodynamic optimality, nor that existing designs are uniquely efficient. However, it does suggest that persistent features of agent architecture may reflect pressure to manage overhead rather than to maximize abstract informational measures.

From this perspective, limits on sustained agency are neither mysterious nor purely theoretical. They arise from ordinary physical constraints on maintaining reliable structure in a noisy world.

The analysis above clarifies how thermodynamic ceilings shape the space of physically sustainable agency. At the same time, the scope of these results is deliberately limited. ECHO imposes constraints on feasibility, not prescriptions for behavior or design.

To avoid misinterpretation, we now make explicit what follows from these arguments and what does not. In particular, we delineate which claims are established by the thermodynamic framework presented here, which remain open empirical questions, and which lie entirely outside the scope of the present work.

8 Limitations, Non-Claims, and Open Problems

The results presented in this paper establish thermodynamic constraints on sustained information processing. They do not constitute a theory of agency, cognition, or consciousness, nor do they resolve semantic disagreements between existing frameworks. This section clarifies the scope of the claims made and identifies the open questions required to connect feasibility bounds to empirical systems.

First, ECHO does not translate or equate framework-specific information-theoretic quantities. Quantities such as integrated information, variational free energy, and assembly depth retain their distinct interpretations. ECHO constrains only the physical rates at which such quantities may be reliably maintained when implemented in matter.

Second, the Price-per-Bit Theorem establishes a feasibility bound, not a statement about actual system performance. The architectural inefficiency parameter α_Ω is not derived from first principles here, nor is it claimed to take a universal value. Determining α_Ω for real systems is an empirical and engineering problem that depends on physical architecture, operating conditions, and reliability requirements.

Third, no optimization principle is assumed or implied. Systems are not claimed to maximize integrated information, minimize free energy, or optimize any persistence functional. ECHO constrains what is physically sustainable regardless of how systems behave within those constraints.

Several open problems remain. Most prominently, the operational measurement of information rates and architectural overhead in complex biological and artificial systems requires new experimental and theoretical tools. Establishing how close real systems operate to their thermodynamic ceilings is an open empirical question.

Finally, ECHO does not address normative questions about meaning, value, or purpose. Such interpretations may be layered atop thermodynamic constraints, but they are not entailed by them.

Within these limits, ECHO provides a general and conservative framework for reasoning about the physical feasibility of sustained information processing across diverse theories.

9 Conclusion

This paper introduced ECHO, a thermodynamic constraint framework for reasoning about the physical feasibility of sustained information processing. Rather than proposing a new theory of agency or cognition, ECHO supplies a minimal and interpretation-agnostic envelope within which existing theories must operate.

The core result is the Price-per-Bit Theorem, which establishes an architecture-dependent ceiling on sustained information rates. Reliable maintenance of information under noise requires logical irreversibility, logical irreversibility entails dissipation, and finite dissipation budgets impose hard upper bounds on what can be sustained over time. These bounds hold independently of how information is interpreted or represented.

Applying this constraint across Integrated Information Theory, the Free Energy Principle, and Assembly Theory reveals a shared thermodynamic structure beneath their differences. Apparent theoretical tensions can often be reframed as competition for scarce dissipative resources rather than

as fundamental conceptual incompatibilities. ECHO does not unify these frameworks semantically, but it places them on common physical ground.

A minimal worked example demonstrates that thermodynamic ceilings are not merely formal. Architectural overhead appears immediately at the smallest nontrivial scale capable of sustaining integrated information. The inefficiency parameter α_Ω captures the irreducible gap between what thermodynamics permits in principle and what physical architectures achieve in practice.

The central lesson of ECHO is therefore architectural. Limits on sustained agency arise less from abstract informational capacity than from the cost of enforcing reliable structure in a noisy physical world. No choice of representation, algorithm, or interpretation eliminates this cost; it can only be redistributed across architectures.

ECHO does not answer normative questions or prescribe objectives. Its contribution is prior and more modest: to make explicit the physical constraints that any theory of sustained information must respect. Within those constraints lies the space in which agency, cognition, and construction are possible at all.

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